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BIP Hack Your Device 2026

Digital certificates

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- Basic problem of asymm. cryptology:
 - Where does the public key come from?
 - It is easy to exchange the key
 - Simple means: Confidentiality does not matter, but
 - **Is public key authentic?**

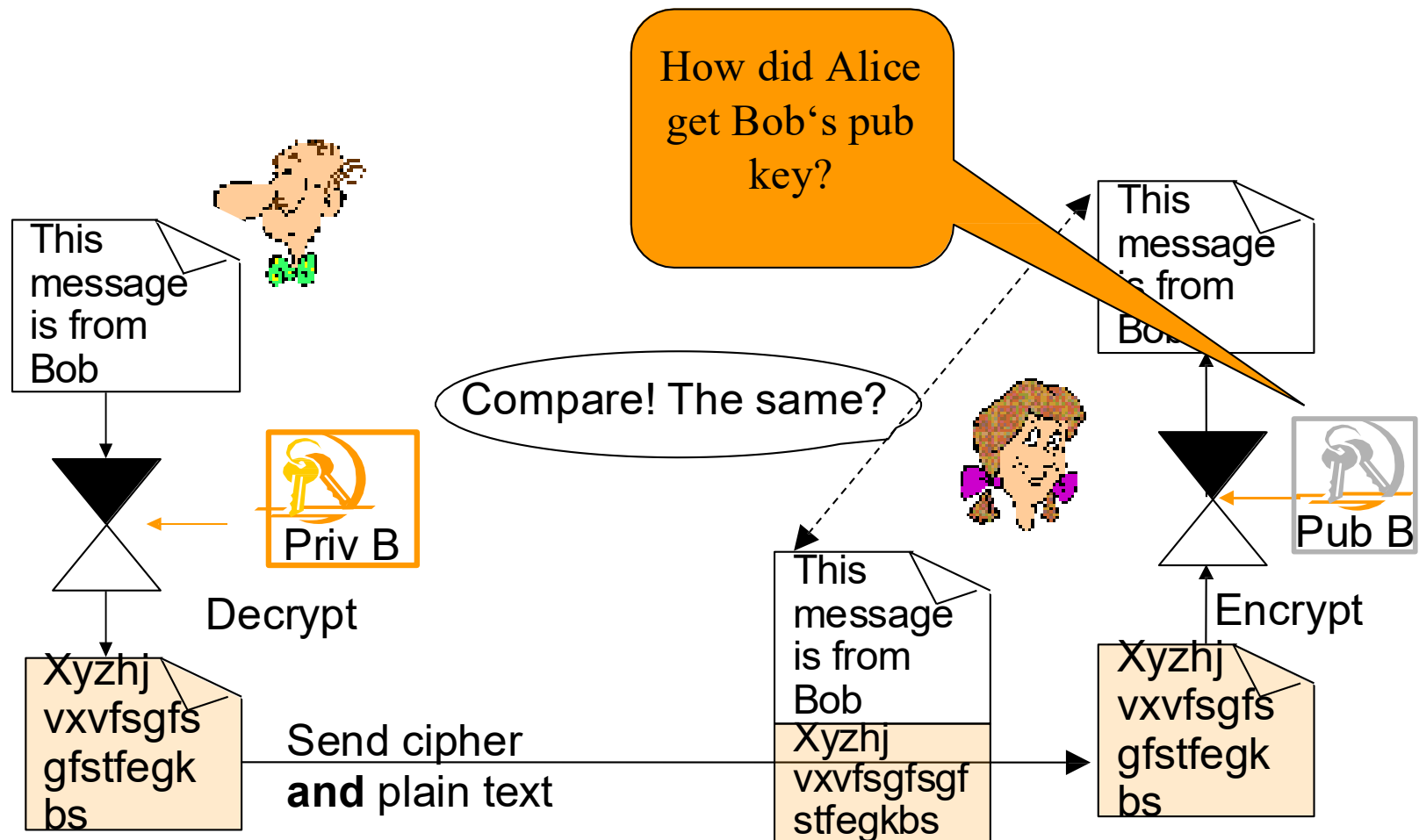
Digital certificates

Digital "signature" recall

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- Asymmetric algorithms for **authenticity** and **integrity** and **non-repudiation**





- Idea:
 - Alice does not ask Bob for public key v. Bob but
- **Certification Authority (CA)**
 - CA manages public keys
 - Sends to Alice the public key v. Bob
 - Attention: **CA** must **authenticate** itself to Alice!
 - How?
 - By means of private key of the CA.
 - Public key of the CA is already stored in the operating system/browser.

This is not
*Digital
Certificate!*
Only idea
about it

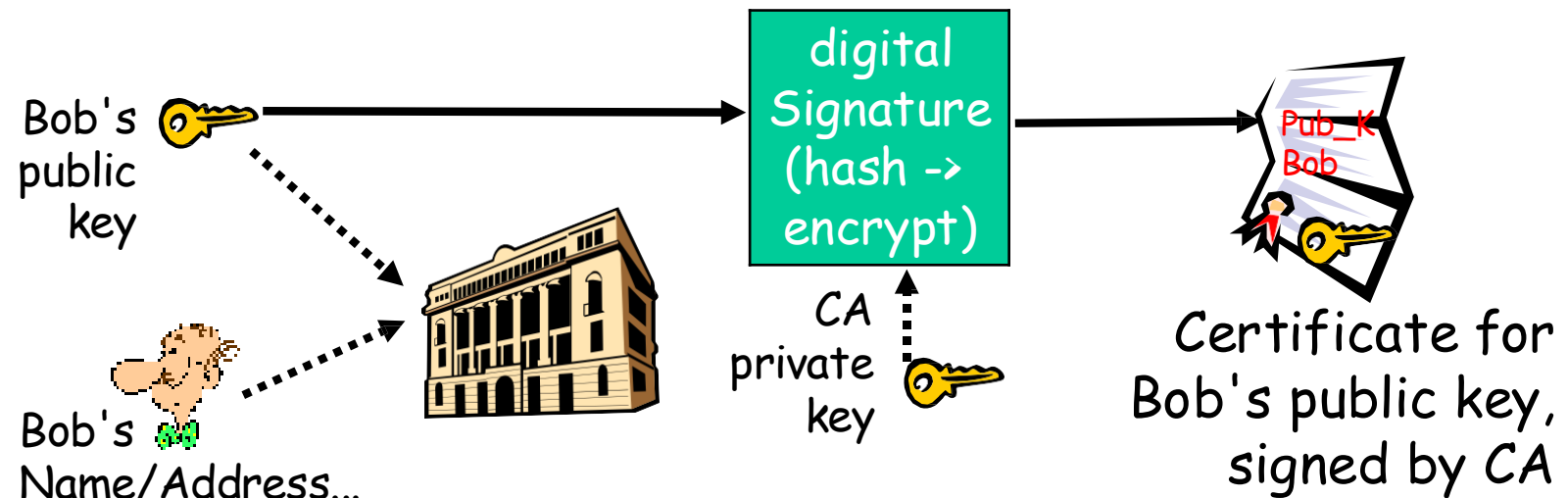


- Improvement of the idea:
 - Instead of asking the CA for pub key_Bob every time
 - The CA **signs** message with its private key
 - Message content
 - "this is Bob's public key"
 - Public key from Bob
 - Bob's Name/Address... = Subject
 - Expiration date
 - This message **plus digital signature** is digital certificate.
- **Certificates bind public key to a subject**
 - Subject is person or organization

Digital certificates Process



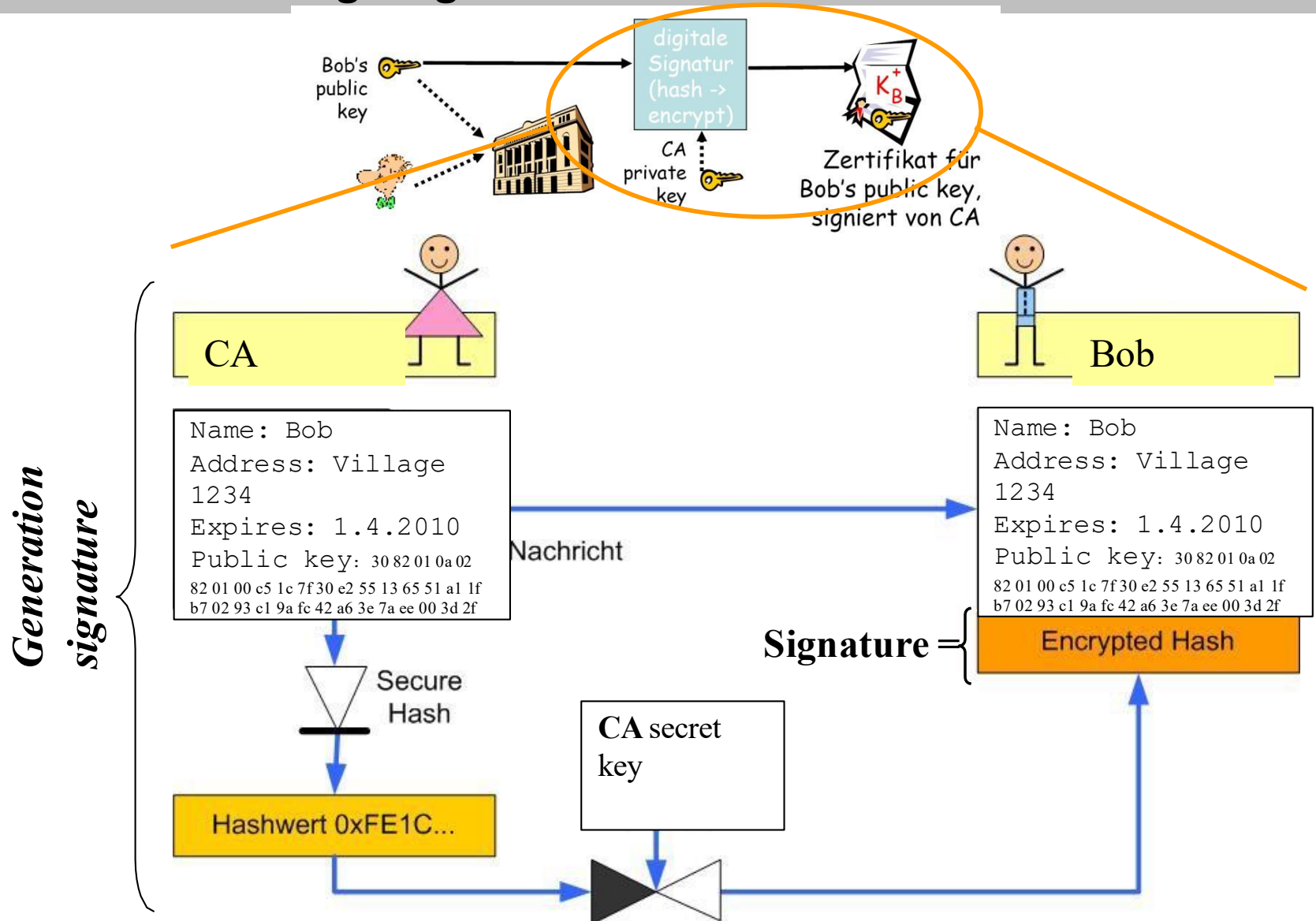
- Bob registers his public key with CA
 - Must prove his identity against CA
- CA generates certificate
 - "this is bob's public key"
 - CA signs message



Digital certificates

Procedure: Signing Details

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Digital certificates

Procedure: Verification

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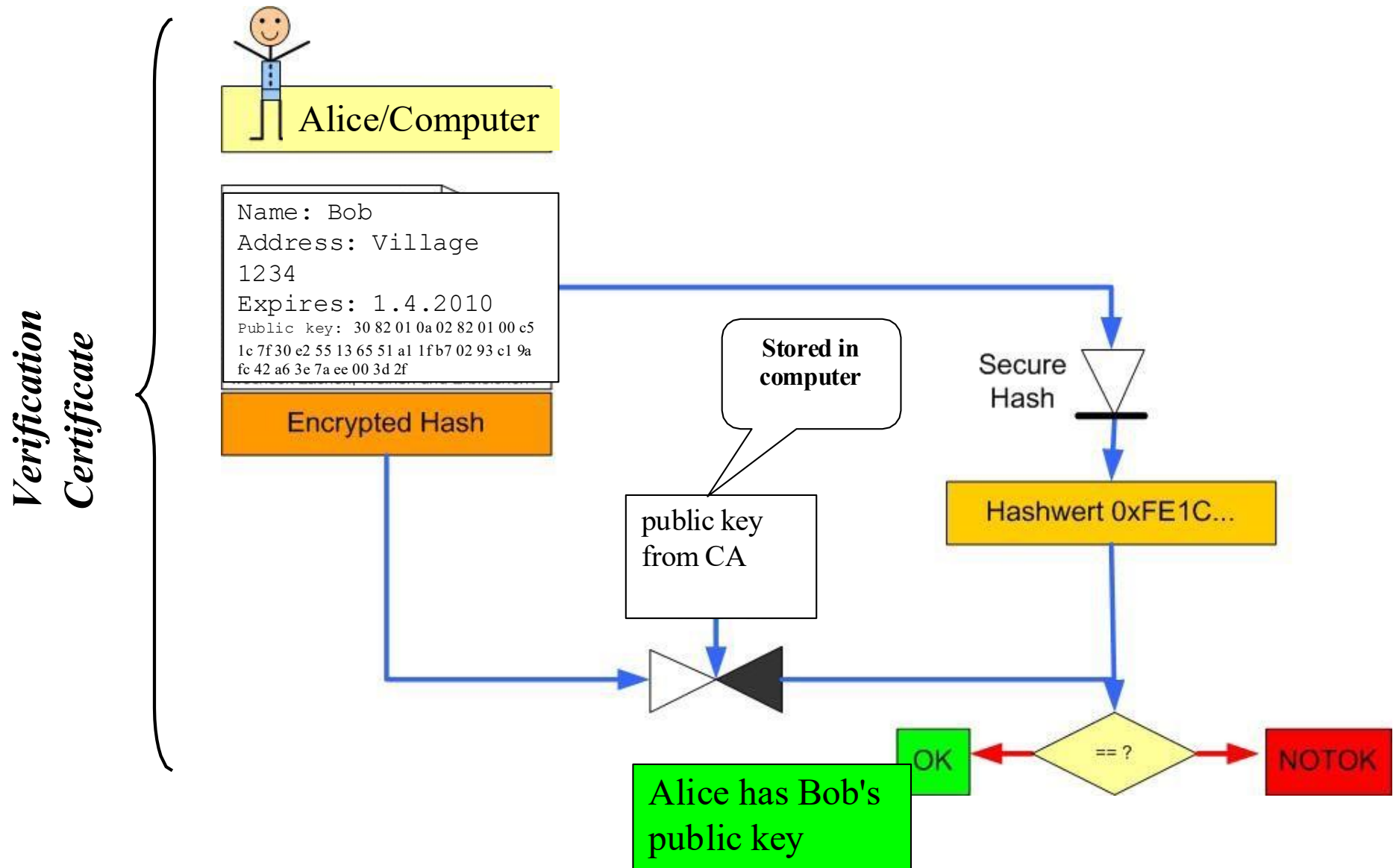


- If Alice wants Bob's public key
- Alice gets Bob's certificate **from** Bob (or elsewhere)
- Alice ('s computer) checks the signature
 - Using the stored **public key of the CA**
 - If ok: Alice knows that public key of Bob is authentic.

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Procedure: Verification

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Applied Crypto

PKI / Management – What is a “PKI”

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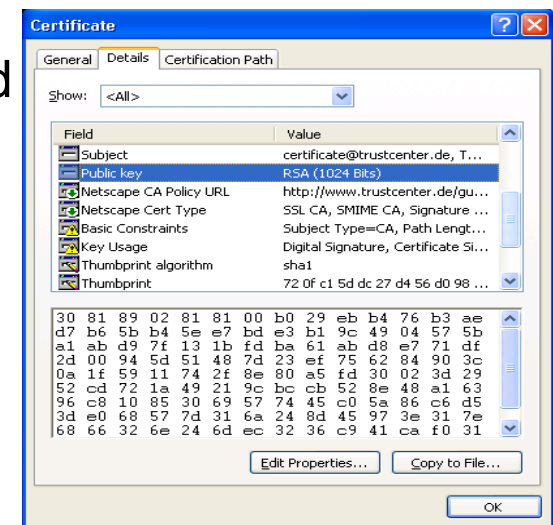


♦ **Public-key infrastructure (PKI)** is a cryptology system that enables issuing, distributing, checking or revocation of digital certificates.

- **Public:** One part of the key is published to everyone needed
- **Key**
- **Infrastructure**

- **Used for:**

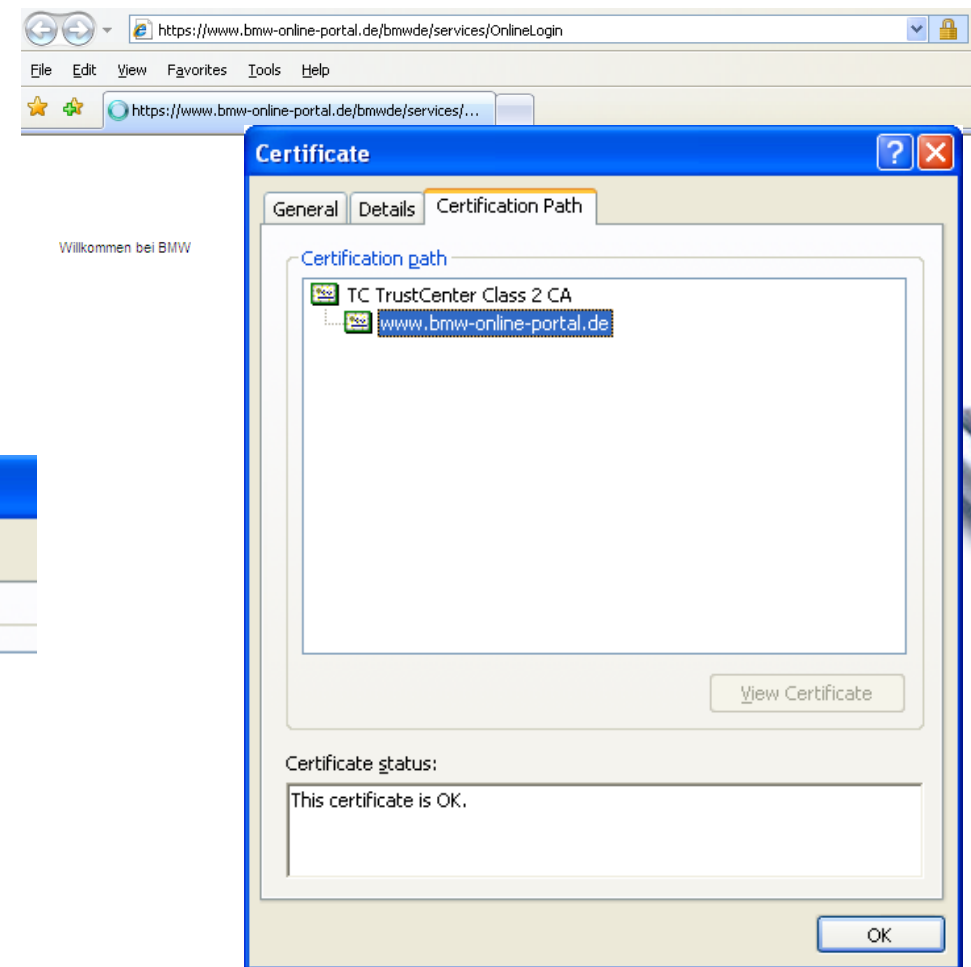
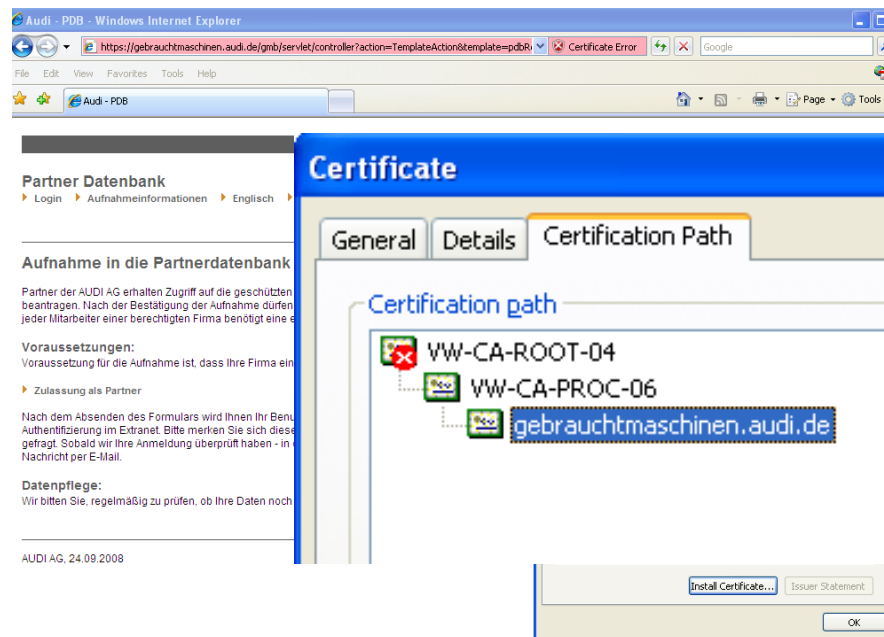
- Generation of cryptographic key pairs
- Generation of digital certificates
- Provision of additional services (eg time stamp)
- Management of digital certificates



PKI / Management – Certification vs Certificate

◆ Trusted (certified) CA

◆ Untrusted (self signed) CA





PKI / Management – Certification vs Certificate

- ♦ What is a PKI **Certification** and what is a Certificate for
 - Certified PKI is based/or signed on/from a CA already in your root stores
(what is a root store? -> will be part of the exercise)
The certificates will be shown as green, trusted or just working . . .
 - “*Self signed*” certificates or issued by an *untrusted* PKI
The certificates will be shown as red, untrusted (yellow arrow) or do not work!



PKI / structure of X-509-v3 certificate

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- ♦ Certificate
- ♦ ...
- ♦ Issuer
- ♦ Validity
- ♦ Certificate holder
- ♦ Certificate holder key information
- ♦ ...
- ♦ Public key of the certificate owner
- ♦ ...
- ♦ **Certificate signature**
 - **Generated by the CA (with their priv key)**